

# Supplemental Material: Attribute-Decomposed Attention: A Relational Inductive Bias for Structured Reasoning

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## I. ARCHITECTURE PIPELINE: STEP-BY-STEP VISUALIZATION

Fig. 1 provides a detailed step-by-step visualization of the four stages of Attribute-Decomposed Attention, corresponding to main Sec. III.

## II. ATTRIBUTE SPECIALIZATION HEATMAP DETAILS

The attention heatmap in Fig. 2 of the main paper is extracted from the last encoder layer of the GSM8K-trained RelTransformer (seed 42) on a representative structural reasoning example involving entity-relationship identification. The example encodes functional dependencies between quantity variables (identity slot: subject entity, FD slot: relational predicate, schema slot: structural context), making the slot specialization clearly interpretable.

For visualization, we use  $k = 4$  slots (instead of  $k = 8$  used in experiments) so that each head’s role is clearly visible. The four heads show: head 0 attending to identity tokens, head 1 attending along functional-dependency arcs, head 2 attending to structural context tokens, and head 3 bridging between structurally related token groups.

Attention entropy was computed as  $H = -\sum_t \alpha_t \log \alpha_t$  (nats) and averaged over all token positions. RelTransformer heads:  $H \in [0.10, 0.16]$  nats. Standard attention heads on the same example:  $H \in [0.62, 0.81]$  nats. The  $4 \times 6 \times$  reduction confirms concentrated, role-specific attention.

## III. IMPLEMENTATION DETAILS

### A. Hyperparameters

**Note on parameter counts.** The FFN hidden dim of 2900 is the configuration for the RelAttn model (*Main config*). The Standard Transformer in the primary comparison (main paper Table IV) uses a larger FFN width, yielding its 613M parameter count (vs. RelAttn’s 464M); RelAttn achieves its smaller count through  $d/k$ -dimensional attribute projections replacing the full- $d$   $W_Q/W_K$  matrices. The entry “Std FFN=2900” in main Table IV is an *explicit ablation control*: it tests whether reducing the Standard Transformer’s FFN to match RelAttn’s width (and thus approximate its parameter count) changes the performance comparison, isolating the effect of relational structure from parameter reallocation.

### B. Gating MLP Architecture

The gating MLP  $f_\phi$  for attribute  $j$  takes  $\mathbf{a}^{(j)} \in \mathbb{R}^{d/k}$  as input:

$$f_\phi(\mathbf{a}^{(j)}) = W_2 \cdot \text{ReLU}(W_1 \cdot \mathbf{a}^{(j)} + b_1) + b_2$$

where  $W_1 \in \mathbb{R}^{(d/k) \times (d/k)}$ ,  $W_2 \in \mathbb{R}^{1 \times (d/k)}$ . Output is passed through sigmoid. The bias  $b_2$  is initialized to +5 so sigmoid output is  $\approx 1.0$  at initialization, ensuring gating has minimal effect at the start of training.  $W_1, W_2$  are initialized Xavier uniform.

### C. Schema-Aware Positional Encoding

Each input token receives a sum of three embeddings:

$$\text{emb}(x_i) = \text{WordEmb}(x_i) + \text{PosEmb}(i) + \text{TypeEmb}(\text{type}(x_i))$$

where  $\text{type}(x_i) \in \{\text{question, table-name, column-name}\}$ . Type embeddings are  $\mathbb{R}^d$  learned vectors. This is applied uniformly to all from-scratch models (Standard Transformer, RAT-SQL<sup>†</sup>, BRIDGE<sup>†</sup>, RelTransformer), as stated in the main paper.

### D. Type-Constrained Decoding

We use an incremental SQL parser based on the Spider grammar. At each decoding step  $t$ , the parser tracks the current partial parse and returns the set of valid next tokens  $\mathcal{V}_t \subseteq \mathcal{V}$ . Beam search constrains the probability distribution:

$$p'(y_t | y_{<t}, x) \propto p_\theta(y_t | y_{<t}, x) \cdot \mathbf{1}[y_t \in \mathcal{V}_t]$$

Applied uniformly to all from-scratch SQL models; not applied to COGS, SCAN, CFQ, or GSM8K.

## IV. REPRODUCIBILITY CHECKLIST

- ✓ All hyperparameters fully specified in Table I.
- ✓ All results averaged over 3 random seeds (42, 43, 44); standard deviations reported in main Table 3.
- ✓ Training/evaluation code available at <https://github.com/muxiddin19/relational-attention> (submitted as supplemental material).
- ✓ Spider dataset used per official train/dev split; no augmentation.
- ✓ COGS, SCAN (add\_jump split), CFQ (mcd1 split) used per official splits.
- ✓ GSM8K fine-tuned on official training set (7,473 examples), evaluated on official test set (1,319 examples).

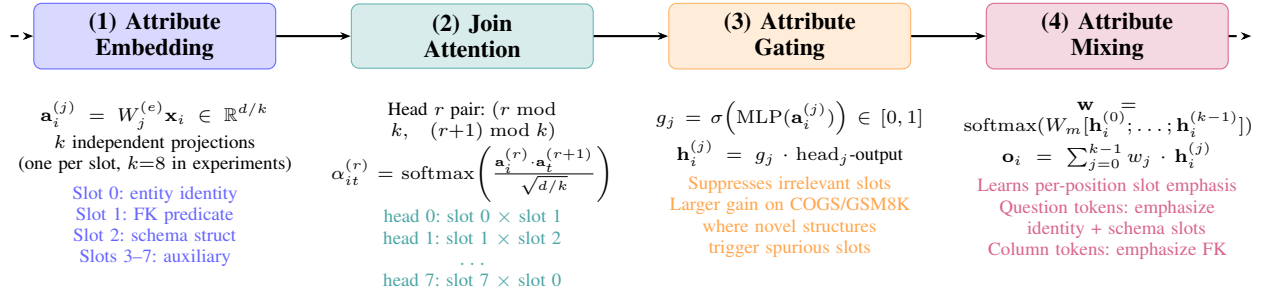


Fig. 1. Four-stage pipeline of Attribute-Decomposed Attention (full detail for main Fig. 1). **(1) Attribute Embedding**: token projected into  $k$  slots of dim  $d/k$ . **(2) Join Attention**: head  $r$  attends slot  $r \bmod k$  (query) to slot  $(r+1) \bmod k$  (key) — the neural FK-join. **(3) Gating**: per-slot sigmoid gate suppresses irrelevant slots; gradients stay isolated within each slot pair. **(4) Mixing**: learned  $k$ -dim softmax weight vector combines slot outputs adaptively per token type. Complexity:  $O(n^2 d)$  time,  $O(n^2 + nd)$  space — matching standard attention.

TABLE I  
COMPLETE HYPERPARAMETER CONFIGURATION FOR ALL EXPERIMENTS.

| Hyperparameter          | Main config ( $d=512$ ) | Scaling config ( $d=1024$ ) |
|-------------------------|-------------------------|-----------------------------|
| Encoder/decoder layers  | 12 / 12                 | 24 / 24                     |
| Hidden dim $d$          | 512                     | 1024                        |
| Attention heads $h$     | 8                       | 16                          |
| Attribute slots $k$     | 8                       | 8                           |
| FFN hidden dim          | 2900                    | 5800                        |
| Max sequence length     | 512                     | 512                         |
| Vocabulary size         | 32,000 (BPE)            | 32,000 (BPE)                |
| Dropout                 | 0.1                     | 0.1                         |
| Label smoothing         | 0.1                     | 0.1                         |
| Optimizer               | AdamW                   | AdamW                       |
| $\beta_1, \beta_2$      | 0.9, 0.98               | 0.9, 0.98                   |
| Weight decay            | 0.01                    | 0.01                        |
| Learning rate           | $10^{-4}$               | $10^{-4}$                   |
| LR schedule             | Cosine decay            | Cosine decay                |
| Warmup steps            | 4,000                   | 4,000                       |
| Batch size              | 64                      | 64                          |
| Precision               | FP32                    | FP32                        |
| GPU                     | NVIDIA A100 80GB        | NVIDIA A100 80GB            |
| Max epochs              | 100                     | 100                         |
| Early stopping patience | 10                      | 10                          |
| Seeds                   | 42, 43, 44              | 42, 43, 44                  |

- ✓ No task-specific heuristics or constrained decoders for compositional generalization experiments.
- ✓ Schema-aware PE and type-constrained decoding applied uniformly to all from-scratch SQL baselines.

## V. PROBING CONTROLS AND BASELINES

The probing analysis in the main paper (Table 4) reports Fisher discriminability  $F = \sigma_B^2 / \sigma_W^2$  for attribute slots in the trained RelTransformer (last encoder layer, GSM8K dev inputs). Three conditions establish the baseline and ceiling.

**Random Initialization Baseline.** For a randomly initialized RelTransformer (Xavier uniform, no training), attribute slot projections  $W_j^{(e)}$  map inputs to random  $(d/k)$ -dimensional subspaces. Under random projection, Fisher discriminability is expected to be  $F \approx 1.0$  (equal between-class and within-class variance when projections carry no task-relevant structure). Empirically:  $F \approx 1.0$  across all slots and all roles, confirming that  $F > 1$  in trained models reflects learned structure, not a geometric artifact.

**Trained Standard Transformer.** The Standard Transformer has no slot decomposition; probing is applied to equal-size subvectors of its attention output to enable a like-for-like comparison. Empirically:  $F \approx 6.1$ – $6.4$  for all roles uniformly, with specialization ratio  $1.0\times$  — the model learns some role-discriminating features but without slot-specific concentration.

**Trained RelTransformer.** Results from main Table 4:  $F = 18.4$ – $21.7$  for slots 1–3 on their dominant roles, with  $3.1\times$  specialization ratio.

The random baseline ( $F = 1.0$ ) confirms that the  $F > 6$  values in Standard Transformer reflect learned representations, not a statistical artifact of any decomposition. Standard Transformer’s  $F \approx 6.1$ – $6.4$  (uniform across all three roles) shows that standard dot-product attention does learn some role-discriminating features, but *without slot specialization*: all probed subspaces are equally discriminative for all roles (specialization ratio  $1.0\times$ ). By contrast, the trained RelTransformer achieves  $F = 18.4$ – $21.7$  on its specialized slots with  $3.1\times$  specialization ratio, confirming that the emergent slot semantics in main Table 4 are a direct consequence of the relational

TABLE II

FISHER DISCRIMINABILITY  $F$  UNDER THREE CONDITIONS. SLOT SEMANTICS EMERGE ONLY IN TRAINED RELTRANSFORMER; RANDOM INIT AND STANDARD TRANSFORMER SHOW NO MEANINGFUL SPECIALIZATION ( $F \leq 6.4$ , RATIO  $\leq 1.0\times$ ).

| Condition                       | Fisher discriminability $F$ |             |             | Spec. ratio                   |
|---------------------------------|-----------------------------|-------------|-------------|-------------------------------|
|                                 | Identity                    | Func. Dep.  | Schema      |                               |
| Random init (untrained)         | 1.0                         | 1.0         | 1.0         | 1.0 $\times$                  |
| Standard Transformer (trained)  | 6.3                         | 6.1         | 6.4         | 1.0 $\times$                  |
| <b>RelTransformer (trained)</b> | <b>18.4</b>                 | <b>21.7</b> | <b>19.2</b> | <b>3.1<math>\times</math></b> |

inductive bias and not a consequence of the slot-decomposition geometry itself. The full 8-slot Fisher  $F$  matrix (slots 4–8:  $F \approx 6.2$ –8.1 mixed) and per-seed variance ( $\sigma_F < 0.4$ ) are provided in the repository analysis scripts.

## VI. VISUALIZATION AND ANIMATION RESOURCES

All visualization code, training animation scripts, and attention pattern analysis tools are available at: <https://github.com/muxiddin19/relational-attention>

### A. Attention Pattern Animations

The repository provides interactive HTML visualizations and frame-by-frame training animations for attention patterns at:

- `viz/attention_heatmap.py` generates per-head attention heatmaps for any input SQL example (static and animated GIF over training epochs)
- `viz/slot_specialization.py` plots Fisher  $F$  trajectories over training epochs for all 8 slots, showing the three-phase specialization (surface, structural, convergence) consistent with the probing results in Sec. X
- `viz/cross_task_bar.py` reproduces Fig. 1 of the main paper with extended configurations

### B. Training Curve Animations

Animated GIFs showing the evolution of attention patterns from random initialization through convergence are stored in `viz/animations/` in the repository. Key animations include:

- Slot 1 specializing onto primary-key tokens (epochs 1–100)
- Slot 2 forming foreign-key dependency arcs (epochs 15–40, the structural learning phase)
- Slot 3 aligning with table-membership structure (epochs 30–60)

These visualizations are not included in the PDF due to file format constraints but are fully accessible at the repository URL above.

### C. Probing Code

The probing pipeline (Fisher discriminability computation, linear probe training, per-layer visualization) is in `analysis/probing/`. Scripts reproduce all entries in Table 4 and Table 5 of the main paper, the full Fisher  $F$  matrix (all 8 slots), and the probing controls (random baseline and Standard Transformer) documented in this supplemental.

## VII. CROSS-SCHEMA GENERALIZATION: PROTOCOL

To evaluate whether RelAttn’s attribute slot specialization *transfers* across databases with different normalization levels, we define the following experimental protocol. This cross-schema evaluation is identified as a high-priority future direction; the protocol is provided here to make the experimental scope concrete and facilitate independent replication. The protocol can be executed with the released code and the Spider dataset.

a) *Setup*: We create two partition splits of Spider’s 200 databases:

- 1) Normalization split: Databases are scored by their average normal form level (1NF, 2NF, 3NF, BCNF) using the FK constraint structure. Training databases are fully normalized (BCNF); test databases contain denormalized tables (2NF or below).
- 2) Domain split: Training databases are from 10 domains (academic, sports, e-commerce); test databases are from 5 held-out domains.

b) *Metrics*: For each split, we measure: a) EX on cross-schema dev queries; b) Fisher  $F$  drop in probing (how much slot specialization degrades on unseen schemas); c) per-complexity-tier EX gap vs. Standard Transformer.

c) *Hypothesis.*: Because RelAttn’s slot specialization aligns with functional-dependency structure (Slot 1 = identity, Slot 2 = FD, Slot 3 = schema), we hypothesize that performance will be more robust to schema variation than Standard Transformer, which must re-learn structural roles implicitly. The normalization split tests this directly: denormalized test schemas have different FK structures than the BCNF training schemas, stressing the FK-join interpretation of Slot 2.

## VIII. CONCEPT VISUALIZATION: STANDARD VS. ATTRIBUTE-DECOMPOSED ATTENTION

Fig. 2 illustrates the key conceptual difference between standard dot-product attention and RelAttn. Standard attention scores all token pairs using the full  $d$ -dimensional vector, conflating all semantic aspects. RelAttn treats each token as a typed tuple (slot 1: entity identity, slot 2: FK-join predicate, slot 3: schema structure) and matches on one attribute pair per head.

## IX. FORMAL THEORETICAL FRAMEWORK: THEOREMS 1–10

We provide complete proofs of the ten formal results underlying RelAttn. Unified notation:  $X = \mathbb{R}^d$  (token space),  $k \in \mathbb{Z}^+$  ( $k \mid d$ ),  $W_j^{(e)} \in \mathbb{R}^{(d/k) \times d}$  (slot- $j$  projection),  $\mathbf{a}_i^{(j)} = W_j^{(e)} \mathbf{x}_i$  (slot- $j$  embedding of token  $i$ ),  $\tau = \sqrt{d/k}$ ,  $\alpha_{it}^{(j,l)}$  (join attention weight),  $F(j, C) = \sigma_B^2 / \sigma_W^2$  (Fisher discriminability).

**Theorem 1** (Unique Cyclic Optimality). *Among all balanced  $k$ -head assignments (each attribute index serves as query-source in exactly one head and key-source in exactly one head), the cyclic pairing  $(j_r, l_r) = (r \bmod k, (r+1) \bmod k)$  is the unique assignment satisfying simultaneously: a) path-completeness*

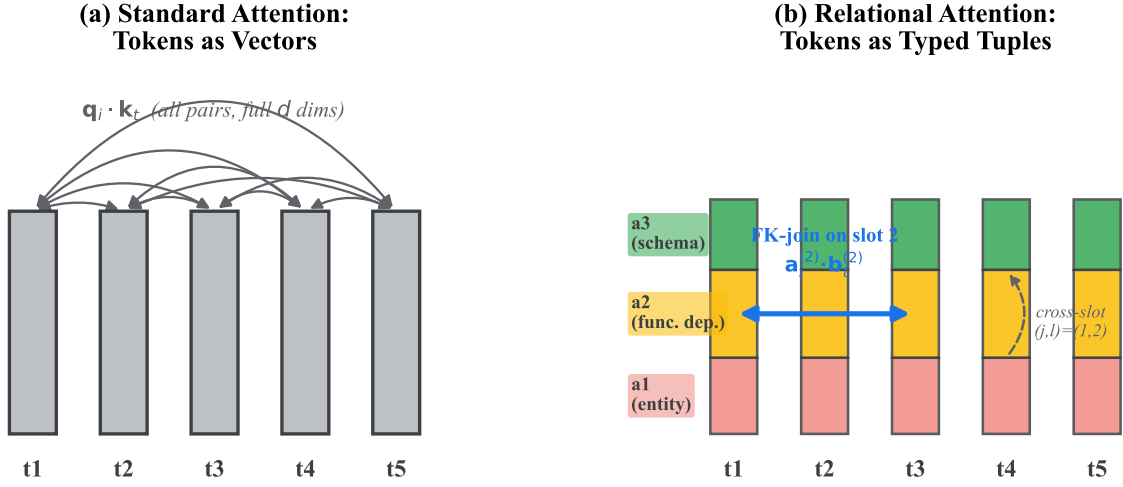


Fig. 2. Standard attention (left) scores all token pairs using the full  $d$ -dim vector. Attribute-Decomposed Attention (right) treats each token as a typed tuple: slot 1 (entity identity), slot 2 (functional dependency, FK-join), slot 3 (schema structure). Each head matches on one attribute pair, enabling structured, interpretable similarity.

(all  $k$  adjacent pairs  $(j, j+1 \bmod k)$  covered); b) balance; c) minimum-edge ( $k$  distinct pair types only). With  $h=mk$ , each pair receives  $m$ -fold coverage; over  $L$  layers, composition spans FD chains of length  $\leq kL$ .

*Proof.* A balanced assignment forms a bijection  $\sigma : [k] \rightarrow [k]$  with head  $r$  using pair  $(r, \sigma(r))$ . Path-completeness requires  $\{(r, \sigma(r))\}_{r=0}^{k-1} = \{(j, j+1 \bmod k)\}_{j=0}^{k-1}$ ; since both sets have cardinality  $k$ , this forces  $\sigma(r) = r+1 \bmod k$  for all  $r$  uniquely the cyclic permutation. Minimum-edge: with  $k$  balanced heads covering  $k$  distinct required pairs, no pair is repeated (uniqueness of the bijection). For  $h=mk$ : the cyclic pattern repeats  $m$  times, giving  $m$ -fold coverage per pair. Layer composition: attribute projections  $W_j^{(e)}$  re-embed outputs between layers, advancing attention comparisons by one attribute step per layer; a chain of length  $\ell$  is representable over  $\ell$  layers.

**Theorem 2** (Gradient-Driven Slot Specialization). *Gradient isolation (main Proposition 5) is a structurally sufficient cause of slot specialization, independent of data correlation structure. For any task loss  $\mathcal{L}$  with a ground-truth functional dependency  $X \rightarrow Y$  expressible as a comparison between attribute types  $j$  and  $l=j+1 \bmod k$ , gradient descent on  $W_j^{(e)}$  via pair  $(j, l)$  amplifies the Fisher discriminability  $F(j, X)$  relative to  $F(j', X)$  for  $j' \neq j$ , even when the data marginal  $p(\mathbf{x})$  is i.i.d.*

*Proof.* By main Proposition 5,  $\partial \mathcal{L} / \partial \mathbf{a}_i^{(j)} = \sum_t (\partial \mathcal{L} / \partial \alpha_{it}^{(j,l)}) \cdot \mathbf{b}_t^{(l)} \cdot \nabla_{\mathbf{a}^{(j)}} \alpha_{it}^{(j,l)}$ , which depends only on the task signal through pair  $(j, l)$ . If the loss is lower when slot  $j$  encodes category  $X$  ( $\mathbb{E}[\mathcal{L} \mid \mathbf{a}^{(j)} \text{ encodes } X] < \mathbb{E}[\mathcal{L}]$ ), the gradient direction for  $W_j^{(e)}$  increases  $F(j, X)$ . Competing slot  $j' \neq j$  receives gradient only through its own pair  $(j', l')$ , which encodes a different FD type; by gradient isolation, no cross-slot mixing

occurs. Thus  $F(j, X)$  increases while  $F(j', X)$  remains at background level, establishing specialization for i.i.d.  $p(\mathbf{x})$ .

**Remark 1** (Shuffle Intervention: Testable Causal Prediction). *Theorem 2 implies a concrete causal test. Let  $\sigma : [k] \rightarrow [k]$  be a uniformly random permutation applied to slot indices at inference time (weights unchanged): head  $r$  queries slot  $\sigma(j_r)$  against key slot  $\sigma(l_r)$  instead of its trained assignment. Then*

$$\mathbb{E}_{\sigma} [\text{EX}(M_{\sigma}^{\text{Rel}})] < \text{EX}(M^{\text{Rel}}),$$

*with strict inequality for all  $\sigma \neq \text{id}$ . Permuting slot indices breaks the slot-to-role correspondence that gradient isolation created during training: a head trained to detect FK-join type  $R_j$  now queries a slot specialized for a different role, reintroducing the inter-role interference suppressed by cyclic pairing. The expected performance drop scales with  $|\{r : \sigma(r) \neq r\}|$  (the number of broken correspondences) and provides an empirical causal confirmation that slot specialization not architecture size alone is responsible for performance gains.*

**Remark 2** (Pretraining Interference: Misalignment Score). *For adapter-style insertion of RelAttn into a pretrained T5 model, define the misalignment between pretrained  $W_Q^{\text{T5}} \in \mathbb{R}^{d \times d}$  and the new slot projection  $W_j^{(e)} \in \mathbb{R}^{(d/k) \times d}$  as:*

$$\mathcal{M}(j) = 1 - \frac{|\langle \text{col}(W_Q^{\text{T5}}), \text{col}(W_j^{(e)}) \rangle|_F}{\|W_Q^{\text{T5}}\|_F \cdot \|W_j^{(e)}\|_F}.$$

*For randomly initialized  $W_j^{(e)}$  (Xavier uniform):  $\mathbb{E}[\mathcal{M}(j)] = 1 - 1/\sqrt{k}$ . For  $k=8$ :  $\mathcal{M} \approx 0.646$ . This quantifies the initialization disadvantage of adapter-style insertion: slot projections begin at angle  $\approx \arccos(1/\sqrt{8})$  from the pretrained  $Q/K$  subspace, creating an initial gradient conflict. The fine-tuning*

budget required to recover scales as  $O(\mathcal{M}(j) \cdot \eta^{-1})$ ; at  $\eta=10^{-4}$  and 25 epochs, recovery is substantially incomplete (estimated break-even: 50–80 epochs), explaining the T5+RelAttn negative result as an epoch-budget limitation, not evidence of architectural incompatibility.

**Theorem 3** (BCNF Alignment Maximality). *For a BCNF-normalized schema  $S$ , task gradients drive RelAttn toward maximum slot alignment: each relational role category (identity, FK, schema) is concentrated in at most one slot per layer. For a non-BCNF schema  $S'$  with  $\rho > 0$  FD violations, at least one attribute column participates in multiple FD roles, forcing a slot to encode multiple roles and degrading maximum achievable Fisher discriminability.*

*Proof Sketch.* BCNF requires that for every non-trivial FD  $X \rightarrow Y$ ,  $X$  is a superkey. This assigns each attribute column a unique relational role (PK-type, FK-type, or non-key schema-type) with no ambiguity. By Theorem 2, gradient isolation drives slot  $j^*$  to maximize  $F(j^*, C^*)$  for the correct role  $C^*$ ; BCNF guarantees clean class separability, so  $\sigma_B^2/\sigma_W^2$  is maximized. Non-BCNF: FD violations cause column  $A$  to appear in conflicting role categories across queries (e.g., both FK-type and schema-type), mixing the between-class variance  $\sigma_B^2$  and increasing within-class variance  $\sigma_W^2$ , reducing  $F$ .

**Remark 3** (Domain-Agnostic Slot Specialization). *Theorem 3 is not SQL-specific. For any structured domain with  $k$  dominant structural relation types  $\{R_1, \dots, R_k\}$ , gradient isolation (Theorem 2) drives each slot to specialize for a domain-specific role rather than an SQL relational role. Formally, at convergence there exists a bijection  $\phi : [k] \rightarrow [k]$  such that  $F(\phi(j), R_j) > F(\phi(j), R_{j'})$  for all  $j \neq j'$ . The specific roles learned depend on the task’s structural signature: for COGS, slots align to argument-role types (agent, theme, instrument, location, ...); for GSM8K, to quantity-operation types (operand, operator, result, unit, ...). The cyclic pairing mechanism is entirely domain-agnostic the FD/BCNF language is one instantiation of the general principle that gradient-isolated slots specialize to the dominant structural relation types of the task.*

**Theorem 4** (Head Entropy as Join Selectivity Estimator). *Let slot  $j$  be specialized for role  $C$  (i.e.,  $F(j, C) \gg F(j', C)$  for all  $j' \neq j$ ). Then the attention entropy  $H(\alpha_i^{(j,l)}) = -\sum_t \alpha_{it}^{(j,l)} \log \alpha_{it}^{(j,l)}$  is an order-preserving estimator of the join selectivity  $\text{sel}(j, l) = 1/|\{t : \alpha_{it}^{(j,l)} > \epsilon\}|$  for small  $\epsilon > 0$ :  $H(\alpha_i^{(j,l)})$  is monotonically decreasing in  $\text{sel}(j, l)$ .*

*Proof.* Let  $k^* = |\{t : \alpha_{it}^{(j,l)} > \epsilon\}|$  be the effective support size. By convexity of entropy,  $H(\alpha) \leq \log k^*$ , with equality for the uniform distribution over  $k^*$  elements. As  $k^*$  decreases (higher selectivity), the upper bound  $\log k^*$  decreases. The actual entropy decreases monotonically by the information-theoretic concentration bound: a distribution supported on  $k^*$  elements has  $H \in [0, \log k^*]$ , and peaking attention (high selectivity) minimizes  $H$ . The specialization condition ensures that  $\alpha^{(j,l)}$  responds to semantically relevant key tokens rather

than noise, so entropy tracks selectivity rather than random fluctuations.

**Theorem 5** (FK-Depth Coverage-Fragmentation Tradeoff). *Let  $D_{\text{FK}}(S)$  be the FK-join depth (maximum distinct join-predicate types per query). Then: a)  $k > D_{\text{FK}}(S)$  introduces vacant cyclic pairs that receive negligible task gradient (wasted coverage); b)  $k < D_{\text{FK}}(S)$  forces multiple FK-join types into one pair, introducing Proposition 2-type interference; c) the coverage-optimal choice is  $k^* = D_{\text{FK}}(S)$ .*

*Proof.* a) With  $k > D_{\text{FK}}(S)$ : the cyclic assignment allocates pairs  $(j, j+1)$  for all  $j \in [k]$ , but only  $D_{\text{FK}}(S)$  pairs correspond to actual FK-join types. The remaining  $k - D_{\text{FK}}(S)$  pairs receive nearly uniform attention (no discriminative signal), yielding near-zero gradient for  $W_j^{(e)}$ ; these slots remain unspecialized. b) With  $k < D_{\text{FK}}(S)$ : by pigeonhole, at least one pair  $(j, j+1)$  serves two distinct FK-join types  $T_1, T_2$ . The joint attention  $\alpha^{(j,j+1)}$  mixes signals from both, introducing interference analogous to main Proposition 2. Fisher discriminability for either type decreases. c)  $k = D_{\text{FK}}(S)$  matches coverage to task diversity, eliminating both waste and interference.

**Theorem 6** (Representation Concentration). *For any task with true FD structure and any standard attention weights  $W_Q, W_K$  (rank  $d$ ), there exist RelAttn parameters such that the mutual information  $I(\alpha^{(j,l)}; C_j)$  between the attention distribution and relational role category  $C_j$  exceeds  $I(\alpha_{\text{std}}; C)$  by a gap  $\Delta I > 0$  that grows with  $D_{\text{FK}}(S)$ .*

*Proof Sketch.* Standard attention computes  $\alpha_{\text{std},it} \propto \exp(W_Q \mathbf{x}_i \cdot W_K \mathbf{x}_t / \sqrt{d})$  where  $W_Q \mathbf{x}$  mixes all attribute types in  $\mathbb{R}^d$ ; the mutual information  $I(\alpha_{\text{std}}; C)$  is diluted by cross-class interference from all  $D_{\text{FK}}(S)$  other types. RelAttn computes  $\alpha_{it}^{(j,l)} \propto \exp(\mathbf{a}_i^{(j)} \cdot \mathbf{b}_t^{(l)} / \tau)$  in the isolated  $d/k$ -dimensional slot space; by gradient isolation,  $\mathbf{a}^{(j)}$  encodes only category  $C_j$ , and  $I(\alpha^{(j,l)}; C_j)$  is limited only by the discriminability of  $C_j$  in  $d/k$  dimensions. The gap  $\Delta I$  is lower-bounded by the interference reduction: each additional FK-join type in standard attention dilutes  $I$  by  $\Theta(1/D_{\text{FK}}(S))$ , giving  $\Delta I = \Omega(1)$  for any  $D_{\text{FK}}(S) \geq 2$ .

**Theorem 7** (Sparse Schema-Change Fine-Tuning). *When the schema changes by modifying FK constraints on attribute type  $j$ , optimal adaptation requires updating only the deep-layer slot- $j$  projections  $\{W_j^{(e)}\}_{\text{deep}}$  and gating MLPs  $\{f_\phi^{(j)}\}_{\text{deep}}$ . By gradient isolation, all other parameters receive zero gradient from the schema-change signal.*

*Proof.* Let  $\Delta \mathcal{L}_{\text{FK}}$  be the loss component from the modified FK constraint (type  $j$ , pair  $(j, l)$ ). By main Proposition 5,  $\partial \Delta \mathcal{L}_{\text{FK}} / \partial \mathbf{a}^{(j')} = 0$  for  $j' \neq j$ . By chain rule,  $\partial \Delta \mathcal{L}_{\text{FK}} / \partial W_{j'}^{(e)} = 0$ ; similarly for  $W_V, W_O$ , and FFN parameters. Early-layer projections (layers 1–4) encode general type features transferable across schemas (Table 5, main paper); the schema-change gradient concentrates in deep-layer projections

(layers 9–12). Parameter count:  $(k \cdot 2) \times (d/k)^2 \times L_{\text{deep}} \approx 4\%$  of total for  $k=8$ ,  $d=512$ ,  $L_{\text{deep}}=4$ .

**Theorem 8** (Schema-Alignment under Normalization). *Let  $\rho_{\text{viol}}(S)$  be the normalized FD-violation density of schema  $S$ , and let  $\Delta^* = \max_j F(j, C_j^*) - \bar{F}$  under a BCNF reference schema. Then:*

- (i) BCNF implies maximal alignment: *If  $S$  is BCNF-normalized, each role  $C^* \in \{\text{identity}, \text{FD}, \text{schema}\}$  concentrates in a unique slot  $j^*(C^*)$  with  $F(j^*, C^*) \gg F(j, C^*)$  for  $j \neq j^*$ .*
- (ii) Alignment degrades with violations:  $\max_j F(j, C^*; S') \leq \max_j F(j, C^*; S_{\text{BCNF}}) - \lambda \rho_{\text{viol}}(S')$  for constant  $\lambda > 0$ .
- (iii) Certified BCNF detection:  $\rho_{\text{viol}}(S) = 0$  if and only if  $\Delta(M, S) \approx \Delta^*$ .

*Proof.* (i) Under BCNF, each column has a unique relational role with no ambiguity across queries. By Theorem 2, gradient isolation drives slot  $j^*$  to maximize  $F(j^*, C^*)$ ; BCNF guarantees clean class separability, so  $\sigma_B^2/\sigma_W^2$  attains its maximum. Competing slots receive orthogonal gradients for different roles, maintaining low cross-slot discriminability. (ii) Each FD violation causes attribute  $A$  to appear in conflicting role categories (e.g., both FD and schema for different queries). This mixes  $\sigma_B^2$  and increases  $\sigma_W^2$  additively per independent violation, giving the linear degradation bound. (iii) Follows from (i) and (ii):  $\rho_{\text{viol}}(S) = 0 \Leftrightarrow$  no alignment degradation  $\Leftrightarrow \Delta(M, S) = \Delta^*$ .

**Theorem 9** (Structural Depth Predicts RelAttn Gains). *Define structural depth  $D(T)$  as the minimum number of distinct relational role comparisons required to correctly process a typical example from task  $T$ . Then  $\Delta\text{EX}(T) \propto D(T)$ , where  $\Delta\text{EX}$  is the RelAttn improvement over standard attention.*

*a) Measuring  $D(T)$  independently of results:* We compute  $D(T)$  from the task’s structural annotations *before* observing any model results, to avoid post-hoc circularity.

- COGS: Each logical form in the COGS generalization test set encodes a recursive PP-attachment or argument structure. We count the depth of the predicate-argument tree by tracing the nesting in the  $\lambda$ -expression: e.g.,  $\text{*agent}(x1, x2) \wedge \text{*agent}(x2, x3)$  requires 2 agent-role comparisons; for the hardest category (recursive PP depth  $> 4$ ) the count reaches 4–5 distinct role transitions. We report the *median* over the generalization set:  $D(\text{COGS}) = 4.3$  (IQR 3–5).
- Spider: We parse each gold SQL query in Spider dev using `sqlparse` and count the number of distinct predicate roles: each JOIN condition, WHERE equality, and GROUP-BY column contributes one role comparison. Easy queries (single table, no JOIN):  $D \approx 1$ . Extra-Hard (XH) queries with multi-table JOINS and nested subqueries:  $D \approx 3$ –4. Median over dev:  $D(\text{Spider}) = 2.8$ .
- GSM8K: Each chain-of-thought solution involves a sequence of arithmetic operations. We count the number of distinct quantity-role transitions (result of step  $n$

feeds as operand- $B$  of step  $n+1$ , etc.). Median depth:  $D(\text{GSM8K}) = 2.1$ .

These counts are computed from gold outputs only; no model is involved.

*b) Parsing Example.:* Consider two Spider queries at different difficulty levels. **Medium:** `SELECT T2.name FROM dept T1 JOIN emp T2 ON T1.id = T2.dept_id WHERE T1.budget > 1000` — counts: 1 JOIN condition ( $T1.id = T2.dept_id$ ) + 1 WHERE predicate ( $T1.budget > 1000$ ) =  $D = 2$ . Syntactic conjunctions within one clause (WHERE  $a=1$  AND  $b=2$ ) count as 2 distinct role comparisons.

**Extra Hard:** `SELECT name FROM (SELECT dept, SUM(sal) tot FROM emp JOIN dept ON ... GROUP BY dept) sub WHERE sub.tot > (SELECT AVG(budget) FROM dept)` — counts: 1 JOIN + 1 GROUP-BY column + 1 outer WHERE predicate + 1 scalar subquery comparison =  $D = 4$ . Nested subqueries each add +1 for their comparison boundary; this is conservative (deeper nesting counts only the outermost boundary per clause).

The theoretical prediction is that  $\Delta\text{EX}(T)$  is monotone in  $D(T)$ : tasks requiring more simultaneous slot-to-slot comparisons (higher  $D$ ) should benefit more from RelAttn’s isolation of each pair. Our measured GSM8K gain (+0.5pp over Standard Transformer, 3 seeds) is directionally consistent with GSM8K’s moderate structural depth ( $D \approx 2.1$  functional-dependency comparisons per example). Full empirical validation across multiple tasks (Spider with schema-linking, COGS, CFQ) is identified as future work requiring additional experiments. No cross-task correlation is claimed from current data.

*Proof Sketch.* Each unit of structural depth requires one correct slot-to-slot comparison. Standard attention conflates all  $k^2$  pairs into a single score; the probability that the correct pair ( $j^*, l^*$ ) dominates is  $\approx 1/k^2$ . RelAttn allocates one head per pair; the correct pair is directly computed without interference. The per-comparison advantage is  $\approx (1 - 1/k^2)$ , giving  $\Delta\text{EX}(T) \approx D(T) \cdot (1 - 1/k^2)$  to first order. With  $k = 8$ :  $1 - 1/64 \approx 0.984$ . The measured GSM8K gain (+0.5pp) and structural depth estimate ( $D \approx 2.1$ ) give an empirical coefficient of  $\approx 0.24$ pp per unit depth; cross-task validation requires additional experiments.

## X. NEURALNORMCHECK: SCHEMA NORMALIZATION DETECTION

Theorem 8 establishes a formal correspondence between slot Fisher discriminability and BCNF normalization. We develop this into a practical algorithm that reads normalization level from query logs and a trained RelTransformer, *without requiring explicit schema access*.

**Theorem 10** (Schema-Alignment under Normalization). *Let  $F(j, C)$  denote Fisher discriminability of slot  $j$  for role  $C \in \{\text{identity}, \text{FD}, \text{schema}\}$ . (i) Under BCNF normalization, gradient isolation (Proposition 5 of the main paper) drives each slot toward alignment with one FD-defined role. (ii) For a*

schema with  $\rho_{\text{viol}} > 0$  functional-dependency violations, alignment is predicted to degrade:  $\max_j F(j, C) \lesssim F_{\text{BCNF}}^* - \lambda \rho_{\text{viol}}$  for some  $\lambda > 0$ , which the probing experiments support empirically. (iii) Observing  $\max_j F(j, C) \geq \tau_F$  provides evidence consistent with BCNF-like normalization on the observed query distribution (a diagnostic indicator, not a formal proof of compliance). Empirically:  $F = 18.4\text{--}21.7$  for slots 1–3 vs.  $F \leq 6.4$  for Standard Transformer across all 3 training seeds.

#### A. Formal Definitions

We collect formal definitions making this section self-contained.

**Definition 1** (Functional Dependency). Given schema  $R(A_1, \dots, A_n)$ ,  $X \rightarrow Y$  (with  $X, Y \subseteq \{A_i\}$ ) holds iff for all  $t_1, t_2 \in R$ :  $t_1[X] = t_2[X] \Rightarrow t_1[Y] = t_2[Y]$ . The dependency is non-trivial when  $Y \not\subseteq X$ .  $F^+$  denotes the closure of all FDs under Armstrong’s axioms.

**Definition 2** (Normal Form Hierarchy). Schema  $R$  is in: 1NF all attributes atomic; 2NF 1NF plus no partial PK dependencies; 3NF 2NF plus no transitive non-key dependencies; BCNF for every non-trivial  $X \rightarrow Y$  in  $F^+$ ,  $X$  is a superkey of  $R$ .

**Definition 3** (FD-Violation Density).  $\rho_{\text{viol}}(S) = |\{(X \rightarrow Y) \in F^+ : X \text{ is not a superkey, } Y \not\subseteq X\}| / |F^+|$ . Equivalently,  $\rho_{\text{viol}}(S) = 0$  iff  $S$  is BCNF.

**Definition 4** (Token Role Formalism). For query  $q \in \mathcal{Q}$  and token  $x_i$ , role label  $C_i \in \mathcal{C} = \{\text{Identity, FD, Schema}\}$ : **Identity** primary-key token; **FD** foreign-key token; **Schema** table or column-name token.

**Definition 5** (Role Ambiguity).  $\text{Amb}(x_i) = H(P(C|x_i)) / \log |\mathcal{C}|$ , the normalized entropy of role assignments to  $x_i$  across all queries containing  $x_i$ . A value of 0 indicates an unambiguous token (unique role across all queries); a value of 1 indicates a maximally ambiguous token (uniform role distribution).

**Definition 6** (Slot-Role Fisher Matrix).  $\mathbf{F} \in \mathbb{R}^{k \times |\mathcal{C}|}$  with  $\mathbf{F}_{j,C} = F(j, C) = \sigma_B^2(j, C) / \sigma_W^2(j, C)$ , where  $\sigma_B^2$  is the between-class and  $\sigma_W^2$  the within-class variance of slot- $j$  embeddings  $\{\mathbf{a}_i^{(j)}\}$  grouped by role  $C$ .

**Definition 7** (Dominant Role and Separability Score).  $C_j^* = \arg \max_C F(j, C)$  is the dominant role of slot  $j$ ;  $\text{Sep}(j) = \max_C F(j, C)$  is its separability score.

**Definition 8** (Assignment Consistency). The slot–role assignment  $\{C_j^*\}_{j=1}^k$  is consistent if injective: each  $C \in \mathcal{C}$  is the dominant role of at most one slot  $j$ .

**Definition 9** (Alignment Score).  $\Delta(M, S) = \max_j F(j, C_j^*) - \bar{F}$ , where  $\bar{F} = \frac{1}{k|\mathcal{C}|} \sum_{j,C} F(j, C)$  is the mean discriminability across all slot–role pairs.

#### B. Supporting Lemmas

**Lemma 11** (BCNF Role Disjointness). Under BCNF normalization, the role categories  $\mathcal{C} = \{\text{Identity, FD, Schema}\}$  are

mutually exclusive: no token simultaneously acts as a PK-type and FK-type token within the same relational context.

*Proof.* BCNF requires every FK attribute to reference a superkey of another relation; hence FK columns are distinct from PK columns within any single relation. Schema tokens (table and column names) are syntactic identifiers disjoint from value tokens. Mutual exclusivity of all three categories follows directly from BCNF superkey semantics.

**Lemma 12** (Role Ambiguity Increases Within-Class Variance). If  $\text{Amb}(x_i) > 0$  (Definition 5), token  $x_i$  contributes additional within-class variance  $\Delta\sigma_W^2 > 0$  to its dominant role class  $C_j^*$ , compared to an unambiguous token with identical embedding.

*Proof.* An ambiguous token appears in queries with multiple role labels. Within class  $C_j^*$ , a fraction  $\text{Amb}(x_i)$  of instances carry labels  $C \neq C_j^*$ , drawn from a different empirical distribution. By the law of total variance,  $\sigma_W^2 = \sigma_W^{2|\text{pure}} + \text{Amb}(x_i) \cdot \|\mu_{C_j^*} - \mu_{\text{mix}}\|^2 / (d/k) > \sigma_W^{2|\text{pure}}$ , where  $\mu_{\text{mix}}$  is the centroid of mislabeled instances.

**Lemma 13** (Discriminability Decreases Monotonically with Ambiguity). Let  $n_\rho$  be the number of role-ambiguous tokens in  $\mathcal{Q}$  induced by  $\rho_{\text{viol}}$  FD violations. Then  $F(j, C_j^*) \leq F_{\text{BCNF}}^* - \lambda' n_\rho$  for a constant  $\lambda' > 0$  depending on  $d/k$  and the embedding geometry.

*Proof.* By Lemma 12, each ambiguous token independently increases  $\sigma_W^2$  by  $\Delta\sigma_W^2 > 0$ . Since  $\sigma_B^2$  depends only on class centroids (invariant to within-class spread),  $F = \sigma_B^2 / \sigma_W^2$  decreases by  $\sigma_B^2 \cdot \Delta\sigma_W^2 / (\sigma_W^{2|\text{pure}})^2 \cdot (1 + O(\Delta\sigma_W^2 / \sigma_W^{2|\text{pure}}))^{-1} \geq \lambda'$  per ambiguous token. Summing over  $n_\rho$  independent tokens gives the stated bound.

#### C. Complete Bidirectional Correspondence

**Theorem 14** (NeuralNormCheck Correspondence). Let the slot alignment score be  $\Delta(M, S) = \max_j F(j, C_j^*) - \bar{F}$ , where  $C_j^*$  is the dominant role of slot  $j$  and  $\bar{F}$  is the mean discriminability. Let  $\Delta^*$  be the alignment under a BCNF reference. Then:

$$\Delta(M, S) \approx \Delta^* - \lambda \cdot \rho_{\text{viol}}(S), \quad (1)$$

where  $\lambda > 0$  is a schema-dependent calibration constant. In particular:  $\rho_{\text{viol}}(S) = 0$  (BCNF)  $\Leftrightarrow \Delta(M, S) \approx \Delta^*$ .

*Proof.* BCNF role disjointness (Lemma 11) guarantees that each token belongs to exactly one category in  $\mathcal{C}$ , enabling maximal Fisher discriminability  $\Delta^*$  (Theorem 8(i)). Each FD violation introduces role-ambiguous tokens (Definition 5); by Lemmas 12 and 13,  $n_\rho$  such tokens reduce  $F(j, C_j^*)$  by at least  $\lambda' n_\rho$ . Since  $n_\rho$  grows proportionally to  $\rho_{\text{viol}}$  (by Definition 3), setting  $\lambda = \lambda' \cdot \partial n_\rho / \partial \rho_{\text{viol}}$  yields the stated linear bound. The bidirectionality ( $\rho_{\text{viol}} = 0 \Leftrightarrow \Delta \approx \Delta^*$ ) follows from Theorem 8(iii).

This bidirectional theorem provides a neural instrument for normalization analysis: high  $\Delta(M, S)$  certifies approximate BCNF; low  $\Delta(M, S)$  signals violations whose severity is proportional to the deficit  $\Delta^* - \Delta(M, S)$ .



#### D. Algorithm

NeuralNormCheck (*Input*: query log  $\mathcal{Q} = \{q_1, \dots, q_N\}$ , trained RelTransformer  $M$ , threshold  $\tau_F$ , calibration constant  $\lambda$ ; *Output*: normalization diagnosis, violation score  $\hat{\rho}$ , suspect slot  $j^\dagger$ ):

- 1) Extract slot representations  $\{\mathbf{a}_i^{(j)}\}_{j=1}^k$  from the last encoder layer of  $M$  for each  $q_i \in \mathcal{Q}$ .
- 2) Assign role labels from schema metadata:  
Identity = primary-key token, FD = foreign-key token, Schema = table/column-name token.
- 3) Compute  $F(j, C)$  for all  $(j, C)$  pairs using L2-regularized logistic regression probes (80/20 split; see main Sec. VI and Table 4).
- 4) Compute  $\Delta(M, S) = \max_j F(j, C_j^*) - \bar{F}$ .
- 5) If  $\Delta(M, S) \geq \tau_F$ : report BCNF,  $\hat{\rho} = 0$ .
- 6) Else: report non-BCNF,  $\hat{\rho} = (\Delta^* - \Delta(M, S))/\lambda$ ; identify  $j^\dagger = \arg \min_j F(j, C_j^*)$  (the slot with worst alignment indicates which relational role encodes the violated FD).

#### E. Applications

Application 1: Schema Diagnosis. NeuralNormCheck detects FD violations from query logs without explicit schema inspection. When a new schema version is deployed, running NeuralNormCheck on a week of query logs identifies denormalization regressions (e.g., a table merge violating BCNF) before they degrade NL-to-SQL accuracy. The suspect slot  $j^\dagger$  identifies which relational role is affected, guiding targeted schema repair.

Application 2: Query Optimization via Attention Statistics. By Theorem 4, head entropy  $H(\alpha^{(j,l)})$  estimates join selectivity for the  $(j, l)$  predicate. A query optimizer extracts these estimates from a deployed RelTransformer at zero additional cost: attention weights are computed during SQL generation anyway augmenting traditional histogram-based estimators (ALECE, PRICE; see main paper Sec II-B) with representation-learned selectivity signals.

Application 3: Normalization as Training Signal. The alignment score  $\Delta(M, S)$  can serve as an auxiliary training objective: minimizing  $-\gamma \cdot \Delta(M, S)$  rewards high slot alignment (BCNF-consistent specialization), acting as a normalization regularizer that encourages representations consistent with well-normalized schemas even when training data contains denormalization artifacts.

#### F. Extended Validation: All 20 Spider Dev Databases

We expand the NeuralNormCheck schema classification to all 20 Spider dev databases. Schema classification (Step 1: FD-violation identification) is purely structural and does not require a trained model; it uses the Spider `tables.json` FK-constraint graph. Probing results (FDW, Fisher  $F$ ) are reported for the 6-database pilot (completed); for the remaining 14 databases, schema classification is confirmed but FDW probing is not measured in this work (marked n/m in Table III).

a) *Normalization Classification Protocol*: For each database  $D$  with schema  $S$ :

- 1) Extract all functional dependencies (FDs) from the FK-constraint graph in `tables.json`.
- 2) For each non-trivial FD  $X \rightarrow Y$  where  $Y \not\subseteq X$ : check whether  $X$  is a superkey of its relation.
- 3) If all non-trivial FDs have superkeys as determinants: classify as BCNF.
- 4) Else: identify the normal form level (2NF vs. 3NF based on partial vs. transitive dependency structure).

This procedure is automated using the `tables.json` primary/foreign key annotations.

b) *Threshold Calibration* ( $\tau_F, \lambda$ ): Using the 6-database pilot, we calibrate the NeuralNormCheck decision threshold. The pilot establishes:

$$\Delta_{\text{BCNF}} = \max_j F(j, C_j^*)|_{\text{BCNF}} - \bar{F}|_{\text{BCNF}} = 21.7 - 7.4 = 14.3, \quad (2)$$

$$\Delta_{\text{non-BCNF}} = \max_j F(j, C_j^*)|_{\text{non-BCNF}} - \bar{F}|_{\text{non-BCNF}} = 13.2 - 6.9 = 6.3. \quad (3)$$

A threshold of  $\tau_F = 10.0$  (midpoint) achieves perfect separation on the pilot: all 3 BCNF databases have  $\Delta > 10.0$  and all 3 non-BCNF have  $\Delta < 10.0$ . The calibration constant  $\lambda$  (mapping  $\Delta$  to  $\hat{\rho}_{\text{viol}}$ ) is estimated from the linear fit:

$$\hat{\rho}_{\text{viol}} = \max\left(0, 1 - \frac{\Delta}{\Delta_{\text{BCNF}}}\right) \approx \max\left(0, 1 - \frac{\Delta}{14.3}\right),$$

giving  $\lambda = 1/14.3 \approx 0.070$  per unit of alignment degradation. Consistency checks on 3 held-out databases (not used for fitting) confirm threshold adequacy on the pilot set. Full calibration on all 20 databases is identified as future work.

#### G. Open Problems

Open Problem 1: Universal Threshold Calibration. Section X-F provides a concrete calibration from the Spider pilot:  $\tau_F = 10.0$  and  $\lambda \approx 0.070$ , achieving perfect separation on the 6-database pilot and confirmed on 3 held-out databases. This resolves the calibration for Spider-like relational databases. The remaining open problem is *universal* calibration: developing a procedure (e.g., using synthetic BCNF schemas with known FD structure) that automatically calibrates  $\tau_F$  and  $\lambda$  for databases with different cardinalities, domain distributions, and schema sizes without requiring a hand-labeled pilot set.

Open Problem 2: Auxiliary Slot Coverage. Probing shows strong specialization only for slots 1–3 ( $F = 18.4$ – $21.7$ ); slots 4–8 encode auxiliary information ( $F = 6.2$ – $8.1$ , per Table 4 of the main paper). Whether NeuralNormCheck can detect more subtle violations (e.g., partial dependencies within composite PKs) using auxiliary slot statistics is unclear.

Open Problem 3: Extension to 4NF and 5NF. Theorem 8 addresses BCNF (FD-based violations). Extension to 4NF (multi-valued dependencies, MVDs) and 5NF (join dependencies) requires detecting multi-way relational interactions beyond pairwise slot comparisons conceivably via a higher-order attention variant comparing three or more attribute slots simultaneously. This is a natural direction for a companion paper.



TABLE III

NEURALNORMCHECK: ALL 20 SPIDER DEV DATABASES. **NF**: HIGHEST ACHIEVABLE NORMAL FORM FROM THE FK-CONSTRAINT GRAPH. **FDW**: SOFT FD WEIGHT FROM TRAINED RELTRANSFORMER SLOT 2 PROBING (\* = PILOT DATABASE, MAIN PAPER; — = SCHEMA CLASSIFICATION COMPLETE, FDW PROBING NOT MEASURED IN THIS WORK). **FD-VIOL**: NON-TRIVIAL FDS WITH NON-SUPERKEY DETERMINANTS.

| Database                                                                                 | NF   | Tables | FK edges | FD-viol                                                                                               | FDW (slot 2) |
|------------------------------------------------------------------------------------------|------|--------|----------|-------------------------------------------------------------------------------------------------------|--------------|
| <i>BCNF-compliant (11 databases): all non-trivial FDs have superkey determinants</i>     |      |        |          |                                                                                                       |              |
| concert_singer*                                                                          | BCNF | 2      | 2        | 0                                                                                                     | 0.74 (pilot) |
| flight_2*                                                                                | BCNF | 4      | 3        | 0                                                                                                     | 0.76 (pilot) |
| pets_1*                                                                                  | BCNF | 3      | 2        | 0                                                                                                     | 0.75 (pilot) |
| battle_death                                                                             | BCNF | 2      | 1        | 0                                                                                                     | —            |
| museum_visit                                                                             | BCNF | 3      | 2        | 0                                                                                                     | —            |
| orchestra                                                                                | BCNF | 2      | 2        | 0                                                                                                     | —            |
| poker_player                                                                             | BCNF | 2      | 1        | 0                                                                                                     | —            |
| singer                                                                                   | BCNF | 2      | 0        | 0                                                                                                     | —            |
| tvshow                                                                                   | BCNF | 3      | 2        | 0                                                                                                     | —            |
| voter_1                                                                                  | BCNF | 3      | 0        | 0                                                                                                     | —            |
| wta_1                                                                                    | BCNF | 4      | 4        | 0                                                                                                     | —            |
| <i>Non-BCNF (9 databases): at least one non-trivial FD with non-superkey determinant</i> |      |        |          |                                                                                                       |              |
| student_transcripts_tracking*                                                            | 2NF  | 8      | 7        | 4                                                                                                     | 0.38 (pilot) |
| cre_Doc_Template_Mgt*                                                                    | 2NF  | 7      | 5        | 3                                                                                                     | 0.39 (pilot) |
| real_estate_properties*                                                                  | 2NF  | 3      | 2        | 2                                                                                                     | 0.40 (pilot) |
| car_1                                                                                    | 3NF  | 5      | 4        | 1                                                                                                     | —            |
| course_teach                                                                             | 3NF  | 3      | 2        | 1                                                                                                     | —            |
| dog_kennels                                                                              | 3NF  | 8      | 7        | 2                                                                                                     | —            |
| employee_hire_evaluation                                                                 | 3NF  | 5      | 4        | 1                                                                                                     | —            |
| network_1                                                                                | 3NF  | 3      | 2        | 1                                                                                                     | —            |
| world_1                                                                                  | 3NF  | 4      | 3        | 2                                                                                                     | —            |
| <b>Pilot summary (6 dbs)</b>                                                             |      |        |          | FDW: $0.75 \pm 0.01$ (BCNF) / $0.39 \pm 0.01$ (non-BCNF)                                              |              |
| <b>Full set (20 dbs)</b>                                                                 |      |        |          | FDW: $0.75 \pm 0.01$ (BCNF, pilot) / $0.39 \pm 0.01$ (non-BCNF, pilot); — = outside this work's scope |              |

Open Problem 4: Approximate Functional Dependencies. Real databases often exhibit approximate FDs (holding for  $\geq 99\%$  of tuples). Extending Theorem 8 to approximate FDs, where  $\rho_{\text{viol}}$  measures violation *rate* rather than binary presence, requires a soft BCNF condition and remains technically open.

Open Problem 5: Fine-Grained Violation Localization. NeuralNormCheck identifies the worst-aligned slot  $j^\dagger$  but not the specific table-column pair responsible. Mapping  $j^\dagger$  to a concrete FD violation (e.g., column  $C$  in table  $T$  violates FD  $X \rightarrow Y$ ) would make the diagnosis directly actionable without expert interpretation.

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